

Project Demo Planning

Date	2 July, 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	1 Mark

Project Demo Planning

A well-structured demo plan ensures effective presentation of the OptiCrop system by clearly explaining the problem, demonstrating the machine learning model, and showcasing the Flask-based application. The demo highlights how environmental and soil parameters are used to generate intelligent crop recommendations and suitability analysis in real time.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction	Introduce the project, objectives, business problem, and importance of smart agriculture.	5 min	Ashritha Kalaga
2	Dataset & Data Analysis	Explain the dataset, importing libraries, reading the dataset, univariate, bivariate, and multivariate analysis.	8 min	Ashritha Kalaga
3	Data Preprocessing	Explain null value handling, feature engineering, preprocessing, and outlier handling.	7 min	KATAKOTA INDU HASINI
4	Machine Learning Model	Explain Logistic Regression, model evaluation, saving the model, and crop prediction	10 min	Teppala Sagarika
5	Flask Web Application & Live Demo	Demonstrate the HTML pages, Flask backend, and real-time crop recommendation.	10 min	Madhav Dhavala
6	Conclusion & Future Scope	Summarize the project, discuss benefits, limitations, and future improvements.	5 min	Madhav Dhavala

Demo Flow Summary

Step	Activity	Notes
1	Introduction & Problem Statement	Explain the need for smart agriculture and intelligent crop recommendation.
2	Dataset & ML Model	Explain the dataset, preprocessing, exploratory data analysis, and machine learning model.
3	Live Feature Demonstration	Enter soil nutrient and environmental parameters to display the recommended crop in real time.
4	Q&A Session	Answer questions related to the dataset, model selection, Flask application, performance, and future enhancements.